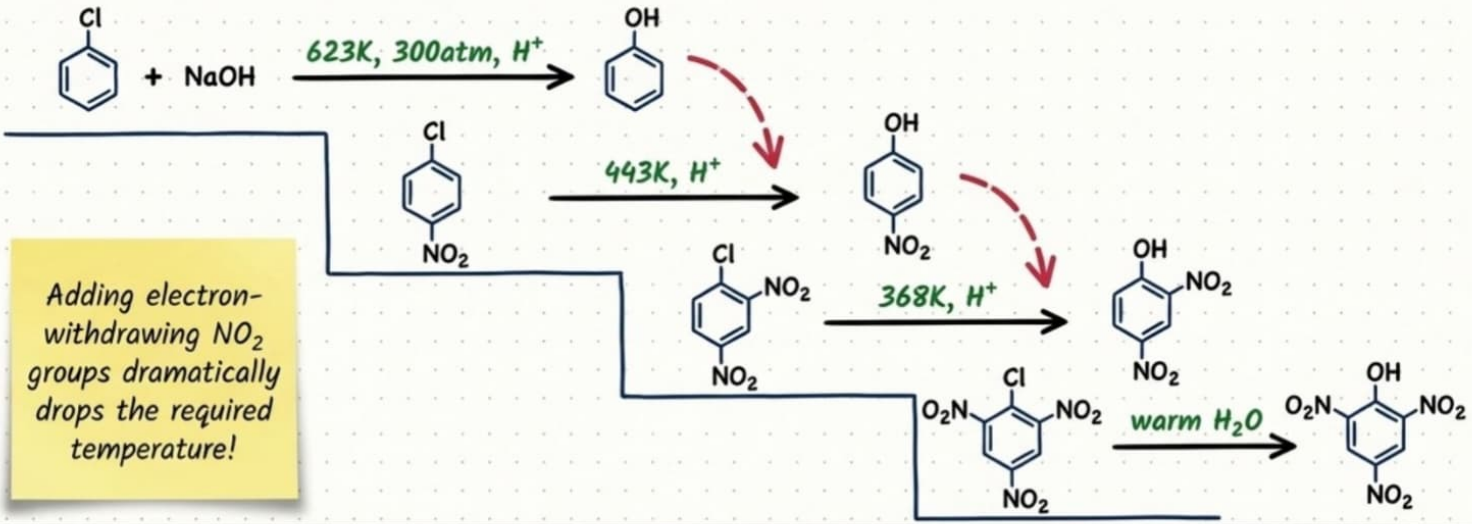
Wurtz Reaction

$$2R-X + 2Na \longrightarrow R-R + 2NaX$$

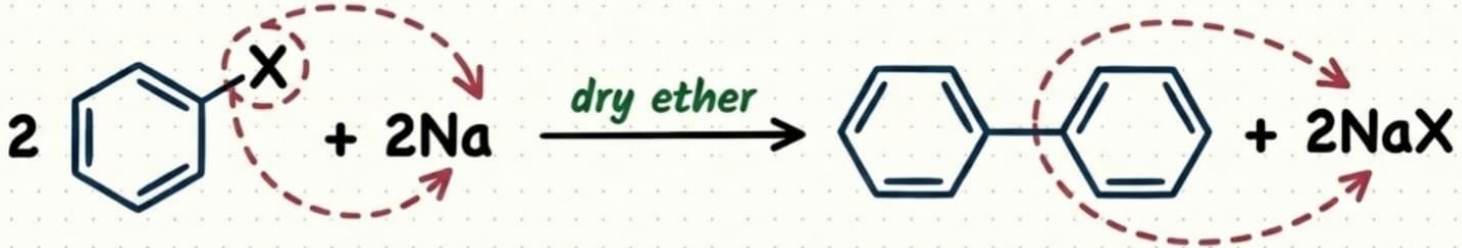
Nucleophilic Substitution of Chlorobenzene (Dow Process)



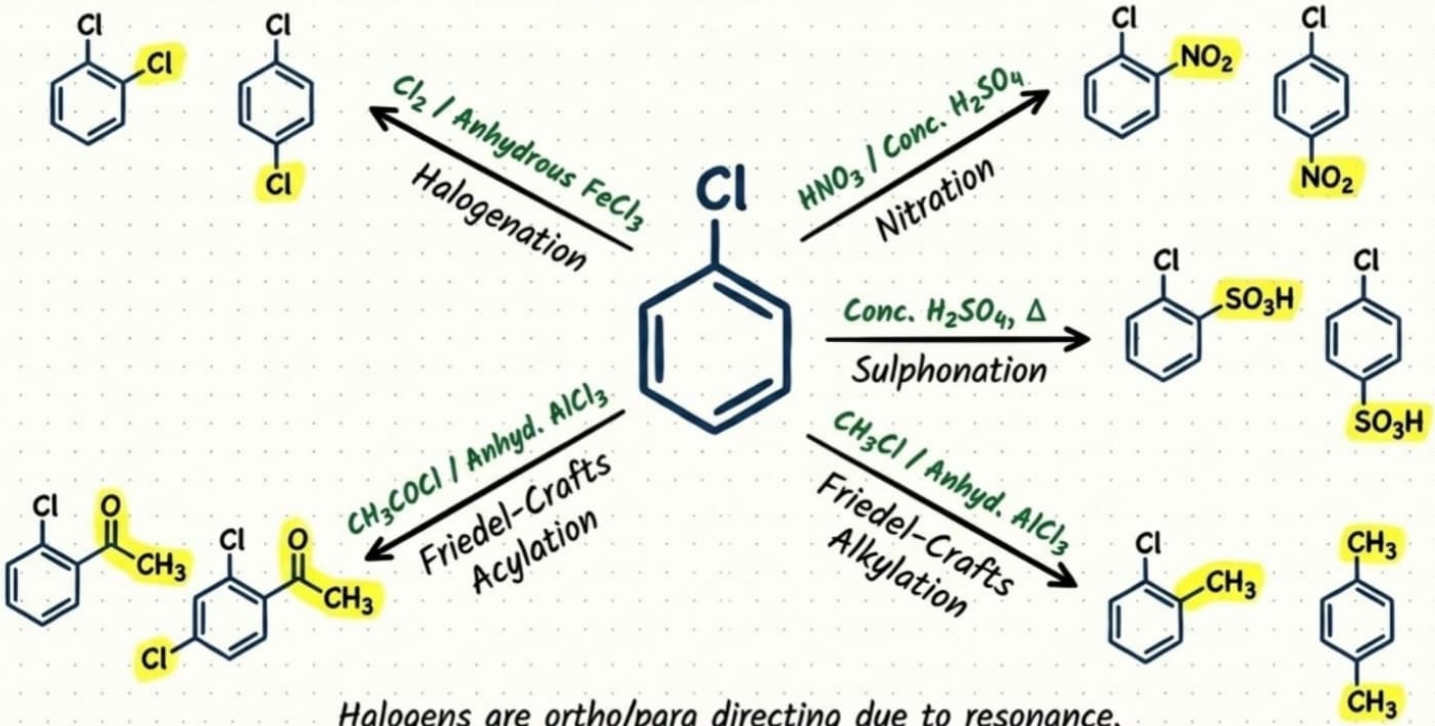
Adding electron-withdrawing NO_2 groups dramatically drops the required temperature!

Wurtz-Fittig Reaction: Aryl + Alkyl

Fittig Reaction: Aryl + Aryl

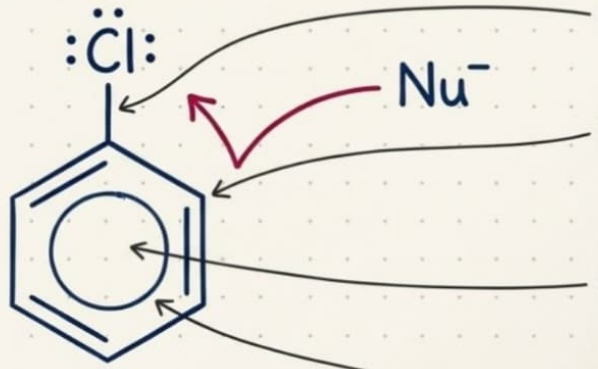


Electrophilic Substitution



Halogens are ortho/para directing due to resonance.

Why are Haloarenes resistant to Nucleophiles?



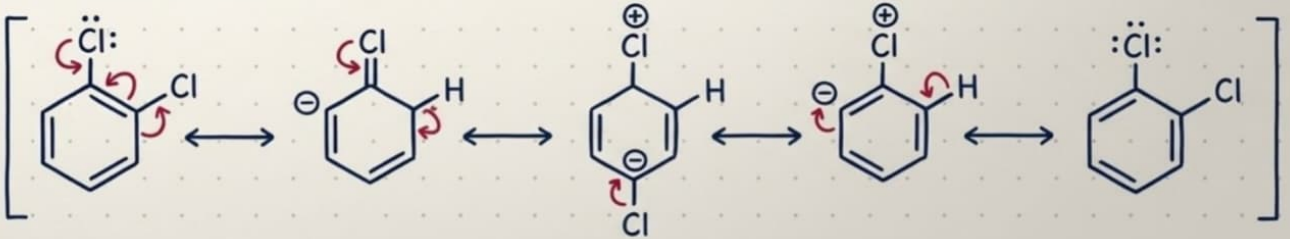
① Resonance: C–X bond develops partial double-bond character; cleavage is difficult.

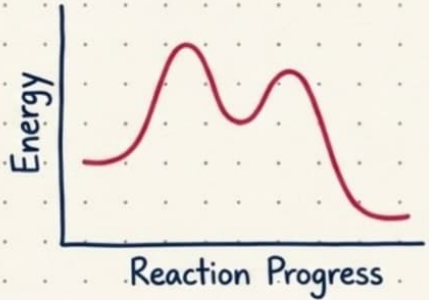
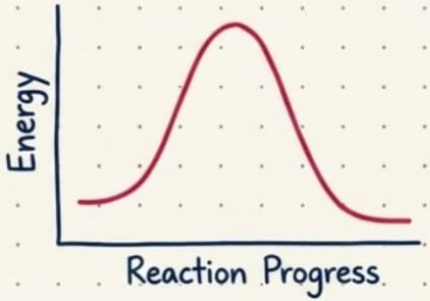
② Hybridisation: sp^2 carbon is more electronegative, holding C–X bond electrons tightly.

③ Instability: A phenyl cation formed would not be stabilised by resonance.

④ Repulsion: Nu^- cannot easily attack due to repulsion from the electron-rich π cloud.

Proof of Resonance:



S_n1 Mechanism	S_n2 Mechanism
2-step reaction (Carbocation forms)	1-step reaction (Transition state forms)
1 molecule drives the reaction	2 molecules drive the reaction
Occurs in polar protic solvent	Occurs in polar aprotic solvent
$I^- > Br^- > Cl^- > F^-$	$I^- > Br^- > Cl^- > F^-$
$3° > 2° > 1°$	$1° > 2° > 3°$
	

Physical Properties: Boiling Points

Trend 1: Halide Size vs Boiling Point

R-F < R-Cl < R-Br < R-I

B.P. increases with increasing halide size (greater van der Waals forces).

Trend 2: Carbon Chain Branching vs Boiling Point

B.P. decreases with branching (decreased surface area).

End of Chapter